

Dispatcher

① Decode state

r_2 r_3 ...
 $d, K^{dec}, \{D_i\}, \tau$

③ Prefill order

r_4 r_5 ...

⑤ Mixed batch

Decode Prefill p^*

Allocator

State

Candidates

Allocation

Candidate chunks

r_4  ...
 r_5  ...
 c_i, H_i

④ Selected chunk

p^*  Later

Each step

② Model + budget

p d K^{dec} A^{pre}

Time cost
model

$\hat{L}$

Time budget

τ Δ D_{target}

Feasible p^*

Time $\hat{L}$ Δ

Tokens d p^* B_{max}